

42. (g) Implementation Plan: B.Tech. in Mechatronics Engineering

Semester - I							Semester - II						
Course Code	Course Title	Hours / Week				C	Course Code	Course Title	Hours / Week				C
		L	T	P	L				T	P			
21LEH101T	Communicative English	2	1	0	3	21LEH102T	Chinese	2	1	0	3		
21MAB101T	Calculus and Linear Algebra	3	1	0	4	21LEH103T	French						
21PYB104J	Physics :Mechanics	3	1	2	5	21LEH104T	German						
21MES102L ¹	Engineering Graphics and Design	0	0	4	2	21LEH105T	Japanese						
21EES101T	Electrical and Electronics Engineering	3	1	0	4	21LEH106T	Korean						
21CYM101T ¹	Environmental Science*	1	0	0	0	21LEH107T	Spanish						
21PDM101L ¹	Professional Skills and Practices	0	0	2	0	21LEH108T	Russian						
21LEM101T ¹	Constitution of India	1	0	0	0	21GNH101J	Philosophy of Engineering	1	0	2	2		
Total Credits 18							21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4	
Semester - III							21CYB101J	Chemistry	3	1	2	5	
Course Code	Course Title	Hours / Week				C	21MHC101P ¹	Elements of Mechatronics systems	2	1	0	3	
		L	T	P	21CSS101J		Programming for Problem Solving	3	0	2	4		
21MAB201T	Transforms and Boundary Value Problems	3	1	0	4	21BTB103T	Biology	2	0	0	2		
21MHC201T	Electrical Actuators and Drives	3	0	0	3	21MES101L ¹	Basic Civil and Mechanical Workshop	0	0	4	2		
21MHC202J	Analog and Digital Electronics	2	0	2	3	21PDM102L ¹	General Aptitude*	0	0	2	0		
21MHC203J	Fluid power system and Automation	2	0	2	3	21GNM101L ¹	Physical and Mental Health using Yoga	0	0	2	0		
21PDH209T ¹	Social Engineering	2	0	0	2	21GNM102L ¹	National Service Scheme						
21MHS201T	Thermodynamics and Heat Transfer	3	0	0	3	21GNM103L ¹	National Cadet Corps						
21MHC204L ¹	Electrical Actuators and Drives Laboratory	0	0	2	1	21GNM104L ¹	National Sports Organization						
21LEM201T ¹	Professional Ethics*	1	0	0	0	Total Credits 25							
21PDM201L ¹	Verbal Reasoning*	0	0	2	0	Semester - IV							
21LEM202T ¹	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3	21MAB202T	Numerical methods	3	1	0	4		
Total Credits 22							21CSC206T	Artificial Intelligence	2	1	0	3	
Semester - V							21MHC205T	Microcontroller and Embedded Systems	3	0	0	3	
Course Code	Course Title	Hours / Week				C	21MHC206T	Mechanics of Solids and Fluids	3	0	0	3	
		L	T	P	21MHC207L ¹		Microcontroller and Embedded Systems Laboratory	0	0	2	1		
21MAB301T	Probability and Statistics	3	1	0	4	21MHC208L ¹	Mechanics of Solids and Fluids Laboratory	0	0	2	1		
21MHC301T	System Dynamics and Control	3	0	0	3	21MHC209T	Project Management and Industrial Practices	2	1	0	3		
21MHC303J	Measurement, Sensors and Interfaces	2	0	2	3	21DCS201P ¹	Design Thinking and Methodology	1	2	0	3		
21MHC302J	Design and Analysis of Machine Elements	2	0	2	3		Professional Elective – I	3	0	0	3		
	Professional Elective – II	3	0	0	3	21PDM202L ¹	Critical and Creative Thinking Skills*	0	0	2	0		
	Open Elective – I	3	0	0	3	Total Credits 24							
21MHC304L ¹	Modelling and Control Laboratory	0	0	2	1	Semester - VI							
21PDM301L ¹	Analytical and Logical Thinking Skills*	0	0	2	0	Course Code	Course Title	Hours / Week				C	
21LEM301T ¹	Indian Art Form	1	0	0	0			L	T	P			
21GNP301L ¹	Community Connect	0	0	2	1	21CSS303T	Data Science	2	0	0	2		
Total Credits 21							21MHC305J	Manufacturing Processes	2	0	2	3	
Semester - VII							21MHC306T	Kinematic Analysis and Dynamics of Mechanisms	3	0	0	3	
Course Code	Course Title	Hours / Week				C	21MHC307P ¹	Model Based Systems Engineering	1	2	0	3	
		L	T	P			Professional Elective – III	3	0	0	3		
21GNH401T	Behavioural Psychology	2	1	0	3	21MHP302L ¹	Project	0	0	6	3		
	Professional Elective – IV	3	0	0	3	21MHP303T ¹	MOOC	3	0	0			
	Professional Elective – V	3	0	0	3	O	Open Elective – II	3	0	0	3		
	Professional Elective – VI				3	21PDM302L ¹	Employability Skills and Practices*	0	0	2	0		
	Professional Elective – VII				3	21LEM302T ¹	Indian Traditional Knowledge	1	0	0	0		
O	Open Elective – III	3	0	0	3	Total Credits 20							
Total Credits 18													

#Students have to register either 21MHP401L or 21MHP402L and 21MHP403L both in eighth semester	Semester - VIII					
	Course Code	Course Title	Hours / Week			C
			L	T	P	
	21MHP401L	Major Project	0	0	30	15
	21MHP402L	Major Project	0	0	20	10
	21MHP403L	Internship#	0	0	10	5
Total Credits						15

